

Linear homogeneous differential equations with constant coefficients (continue)

Given the coefficients $a_1, a_2, \dots, a_n \in \mathbb{R}$, with $n \in \mathbb{N}^*$, we consider the LODE with cc

$$(1) \quad x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x = 0$$

looking for sol. $x = e^{kt}$, $k \in \mathbb{R}$! we found that

$$(2) \quad k^n + a_1 k^{n-1} + \dots + a_{n-1} k + a_n = 0 \quad (\text{called the characteristic eq. of (1)}).$$

it is evident that for $k \in \mathbb{R}$ a root of (2) we have that e^{kt} is a solution of (1).

Q we want to justify now why for the roots of (2) $k_{1,2} = \alpha \pm i\beta$ (with $\alpha, \beta \in \mathbb{R}, \beta \neq 0$) we have that $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$ are solutions of (1).

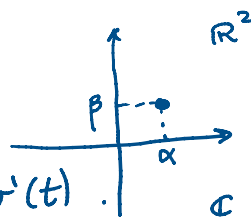
A The complex exponential: $e^{\alpha + i\beta} = e^\alpha (\cos \beta + i \sin \beta)$, $\forall \alpha, \beta \in \mathbb{R}$ Euler's formula

A complex function of real variable: $\gamma: I \rightarrow \mathbb{C}$, where $I \subset \mathbb{R}$ interval.

$$\gamma(t) = u(t) + i v(t), \quad u(t) \stackrel{\text{not}}{=} \operatorname{Re}(\gamma(t)), \quad v(t) \stackrel{\text{not}}{=} \operatorname{Im}(\gamma(t))$$

we have $u, v: I \rightarrow \mathbb{R}$ (real functions)

$$\mathbb{C} \approx \mathbb{R}^2 \quad \alpha + i\beta \in \mathbb{C} \leftrightarrow (\alpha, \beta) \in \mathbb{R}^2$$



When $u, v \in C^1(I)$ we have that $\gamma'(t) = u'(t) + i v'(t)$.

$$\gamma': I \rightarrow \mathbb{C}$$

Proposition 1 Let $u, v \in C^n(\mathbb{R})$. If $\gamma: \mathbb{R} \rightarrow \mathbb{C}$, $\gamma = u + i v$, satisfies eq. (1)

then u and v are solutions of eq. (1). [this prop. is valid also for LODE with varying coeff.]

Proof. $L[\gamma](t) \stackrel{\text{not}}{=} x^{(n)}(t) + a_1 x^{(n-1)}(t) + \dots + a_n x(t)$, $\forall t \in \mathbb{R}$.

$$\left. \begin{aligned} L(\gamma) &= L(u + i v) = \underbrace{L(u)}_{\substack{\uparrow \\ \text{real} \\ L \text{ is linear}}} + i \underbrace{L(v)}_{\substack{\uparrow \\ \text{real}}} \rightarrow \left. \begin{aligned} L(u) &= \operatorname{Re}(L(\gamma)) \\ L(v) &= \operatorname{Im}(L(\gamma)) \end{aligned} \right\} \rightarrow \left. \begin{aligned} L(u) &= 0 \\ L(v) &= 0 \end{aligned} \right\} \Rightarrow \end{aligned}$$

if is that γ satisfies eq. (1) $\Rightarrow L(\gamma) = 0$

$\Rightarrow u$ and v are solutions of (1) (recall that u and v are real functions and $u, v \in C^n(\mathbb{R}, \mathbb{R})$).

Proposition 2 a) $e^{kt} = e^{\alpha t} \cos \beta t + i e^{\alpha t} \sin \beta t$, $\forall t \in \mathbb{R}$

$$b) \quad (e^{kt})' = k e^{kt}, \quad \forall t \in \mathbb{R}$$

$$\boxed{k = \alpha + i\beta, \quad \alpha, \beta \in \mathbb{R}}$$

... not at ... $\alpha t \dots \beta t$

$$b) (e^{zt})' = z e^{zt}, \quad \forall t \in \mathbb{R}$$

Proof. a) use Euler Formula. $u(t) \stackrel{\text{not}}{=} e^{\alpha t} \cos \beta t$, $v(t) = e^{\alpha t} \sin \beta t$ $\alpha, \beta \in \mathbb{R}$
 Note that $u, v \in C^\infty(\mathbb{R})$.

$$b) \left. \begin{aligned} e^{zt} &= u(t) + i v(t), \quad \forall t \in \mathbb{R} \\ (e^{zt})' &= u'(t) + i v'(t) \end{aligned} \right\} \Rightarrow$$

$$u'(t) = \alpha e^{\alpha t} \cos \beta t - \beta e^{\alpha t} \sin \beta t$$

$$v'(t) = \alpha e^{\alpha t} \sin \beta t + \beta e^{\alpha t} \cos \beta t$$

$$\Rightarrow (e^{zt})' = \alpha \cdot e^{zt} + i\beta \cdot e^{zt} = (\alpha + i\beta) e^{zt} = z e^{zt} \quad \square.$$

From here we have:

Let $z \in \mathbb{C}$. e^{zt} satisfies the LHDE (1) iff z is a root of the ch. eq. (2).

1).

Now we apply Prop 1 and Prop 2 a) and obtain that:

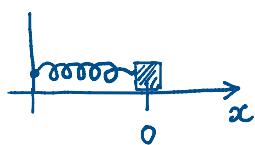
for $z = \alpha + i\beta$, $\beta \neq 0$ root of (2) we get $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$ solutions of (1).

Note that eq (2) has real coefficients, thus if $z_1 = \alpha + i\beta$, $\beta \neq 0$ is a root of (2) then $z_2 = \alpha - i\beta$ is also a root of (2).

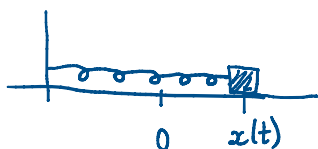
$$e^{(\alpha - i\beta)t} = e^{\alpha t} \cos(-\beta t) + i e^{\alpha t} \sin(-\beta t) = \underbrace{e^{\alpha t} \cos \beta t}_{\text{Re}} + i \underbrace{[-e^{\alpha t} \sin \beta t]}_{\text{Im}}$$

$\Rightarrow e^{\alpha t} \cos \beta t$ and $-e^{\alpha t} \sin \beta t$ are solutions. Nothing new!

An application: the spring-mass system



this is the situation when the system is at an equilibrium position (there is no movement)



$x(t)$ measures the displacement of the mass from the equilibrium position $x=0$.

$x'(t)$ is the velocity, while $x''(t)$ is the acceleration

Newton's second law: $F = m \cdot a$ $m > 0$ $a = x''(t)$

The forces: - the elastic (or restoring) force $F_r = -kx(t)$ with $k > 0$
(Hooke's law)

- the friction (damping) force $F_d = -\gamma \cdot x'(t)$ with $\gamma > 0$

- an external force $F_{ext} = m f(t)$ where f is a given function.

Then $F = F_r + F_d + F_{ext} = -kx - \gamma x' + mf(t)$ $\Rightarrow -kx - \gamma x' + mf(t) = mx'' \Rightarrow$
 $F = ma$

$\Rightarrow \boxed{x'' + \frac{\gamma}{m} x' + \frac{k}{m} x = f(t)}$ second order L non-HDE with constant coefficients
 $\frac{\gamma}{m} > 0, \frac{k}{m} > 0$

we consider the following cases:

Case 1. Motion without damping, without external force

$$x'' + \frac{k}{m} x = 0$$

Case 2. Motion with damping, without external force

$$x'' + \frac{\gamma}{m} x' + \frac{k}{m} x = 0$$

Case 3. Motion without damping, with external force

$$x'' + \frac{k}{m} x = A \cos \omega t, \text{ where } A > 0 \text{ and } \omega > 0 \text{ parameters.}$$

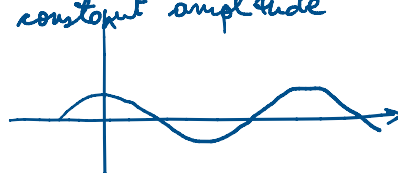
Case 1. $\omega_0^2 = \frac{k}{m}$ $\omega_0 > 0$

$$x'' + \omega_0^2 x = 0 \quad k^2 + \omega_0^2 = 0 \quad k^2 = -\omega_0^2 \quad r_{1,2} = \pm i\omega_0 \mapsto \cos \omega_0 t, \sin \omega_0 t$$

$$x = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t), \quad c_1, c_2 \in \mathbb{R} \text{ arbitrary.}$$

Any function like this is periodic with minimal period $\frac{2\pi}{\omega_0}$,
it oscillates around $x=0$ with constant amplitude
is bounded.

perpetuum mobile (not realistic)



perpetuum mobile (not realistic)

1

Case 2. $x'' + \frac{\gamma}{m} x' + \frac{k}{m} x = 0$

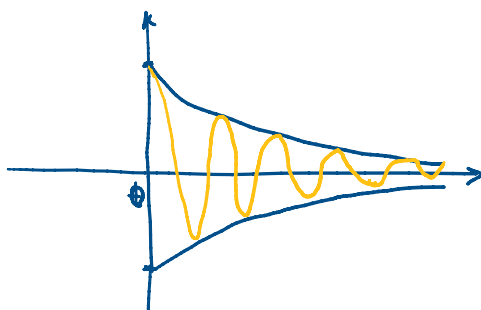
$$r^2 + \frac{\gamma}{m} r + \frac{k}{m} = 0 \quad \Delta = \frac{\gamma^2}{m^2} - 4 \frac{k}{m} = \frac{\gamma^2 - 4km}{m^2}$$

Case 2.1. $\gamma^2 < 4km$ (the damping coeff. is smaller than a critical value $\sqrt{4km}$)

$$r_{1,2} = \frac{-\frac{\gamma}{m} \pm i \frac{\sqrt{4km - \gamma^2}}{m}}{2} = \underbrace{-\frac{\gamma}{2m}}_{\alpha} \pm i \underbrace{\frac{\sqrt{4km - \gamma^2}}{2m}}_{\beta} \mapsto e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t$$

$$x = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t, \quad c_1, c_2 \in \mathbb{R}$$

$e^{\alpha t} \cos \beta t$



oscillates with amplitude that decreases to 0 as $t \rightarrow \infty$.

Case 2.2. $\gamma^2 > 4km$ (the damping coeff. is bigger than $\sqrt{4km}$)
 $r_1, r_2 \in \mathbb{R}, r_1 \neq r_2 \mapsto e^{r_1 t}, e^{r_2 t}$

$$r_1 + r_2 = -\frac{\gamma}{m} < 0, \quad r_1 r_2 = \frac{k}{m} > 0 \Rightarrow r_1 < 0 \text{ and } r_2 < 0$$

$$x = c_1 e^{r_1 t} + c_2 e^{r_2 t}, \quad c_1, c_2 \in \mathbb{R}$$

any such function decreases exponentially to 0 and will not oscillate around 0.

Case 3. $x'' + \omega_0^2 x = A \cos(\omega t)$

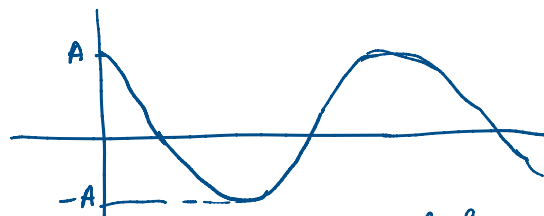
$$x'' + \omega_0^2 x = 0 \quad x_h = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t)$$

$$x_p = ? \quad x = x_h + x_p$$

$$\omega \neq \omega_0 \quad x_p = a \cos \omega t + b \sin \omega t$$

$$a, b \in \mathbb{R} \quad a = ? \quad b = ?$$

$$\omega = \omega_0 \quad x_p = t (a \cos \omega t + b \sin \omega t)$$



the external force.

has period $\frac{2\pi}{\omega}$

and has frequency ω .
 is bounded!

Question: ... is our solution periodic?

is bounded !

Questions: $\omega \neq \omega_0$ is any solution periodic?

$\omega = \omega_0$ any sol. is unbounded resonance.

Tacoma bridge disaster

see report of Alex Pop.

Remark. Resonance is a complex phenomenon. The DE $x'' + \omega_0^2 x = A \cos \omega_0 t$ is just one mathematical model for resonance. It is now known that the mathematical model for the Tacoma bridge disaster is not this simple DE.